

MOUNTAIN VILLAGE, AK, USA

WMO: 703056

Lat: 62.095N

Lon: 163.682W

Elev: 103

StdP: 100.10

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 26635

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.9	(c) -27.4	(d) -32.7	(e) 0.2	(f) -28.1	(g) -31.1	(h) 0.2	(i) -26.3	(j) 17.5	(k) -10.0	(l) 16.0	(m) -7.5	(n) 3.1	(o) 330	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.0	(c) 22.4	(d) 15.1	(e) 19.8	(f) 14.0	(g) 17.7	(h) 13.0	(i) 15.9	(j) 20.7	(k) 14.6	(l) 18.6	(m) 13.6	(n) 16.8	(o) 4.6	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	10.1	16.8	12.8	9.4	15.8	12.0	8.8	14.4	44.8	21.0	41.2	18.4	38.3	17.0	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.7	(b) 12.7	(c) 11.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-14.5	-11.9	-10.8	-4.8	4.2	10.2	12.4	11.3	7.3	0.0	-8.7	-12.0
	DBStd	11.31	8.18	8.30	7.00	6.37	4.85	3.44	3.04	2.66	3.23	4.23	6.03	7.31
	HDD10.0	4347	760	612	644	443	187	38	8	16	88	308	561	681
	HDD18.3	7195	1018	846	902	693	438	246	186	218	331	567	811	939
	CDD10.0	196	0	0	0	0	8	43	83	56	7	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH23.3	32	0	0	0	0	2	7	23	1	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	5.3	5.5	6.0	5.8	5.8	5.2	4.9	4.6	4.7	5.1	5.3	5.0	5.3
Precipitation	PrecAvg	632	44	45	37	32	32	42	61	82	79	65	63	50
	PrecMax	861	92	124	122	67	71	73	112	157	154	109	146	107
	PrecMin	506	9	12	8	7	10	16	24	31	6	12	7	17
	PrecStd	90	22	27	25	16	15	11	21	25	29	24	35	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	2.2	3.0	11.0	21.7	24.1	26.2	22.7	16.5	10.1	3.9	4.2
		MCWB	0.3	1.4	1.1	5.5	12.8	15.4	18.5	15.8	11.5	7.8	3.4	2.1
	2%	DB	0.3	1.0	1.6	8.1	17.0	21.4	23.0	19.4	14.2	8.3	2.4	1.9
		MCWB	-0.2	0.4	0.5	4.0	10.1	14.1	15.8	14.0	10.1	7.0	1.9	1.0
	5%	DB	-0.9	0.1	0.1	5.3	14.0	18.7	20.5	17.7	12.8	7.0	1.2	0.1
		MCWB	-1.4	-0.2	-0.3	2.7	8.5	12.7	15.0	13.3	10.1	6.1	0.8	-0.2
	10%	DB	-2.8	-0.7	-1.1	3.1	11.7	16.4	17.8	15.7	11.7	5.6	-0.1	-1.3
		MCWB	-3.2	-1.0	-1.3	1.3	7.5	11.4	13.7	12.6	9.8	4.9	-0.5	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.6	1.6	5.8	13.5	16.5	19.1	16.4	12.5	8.5	3.6	3.0
		MCDB	0.9	1.8	2.5	10.2	21.5	23.1	25.6	20.7	13.7	9.4	3.7	4.3
	2%	WB	0.0	0.5	0.6	4.3	10.7	14.6	16.6	14.9	11.6	7.4	1.9	0.9
		MCDB	0.3	0.7	1.4	7.8	16.1	19.7	21.4	18.5	12.9	8.1	2.2	1.7
	5%	WB	-1.3	-0.1	-0.2	2.8	9.0	13.2	15.2	13.9	10.6	6.2	0.8	-0.2
		MCDB	-1.0	0.1	0.3	5.1	13.4	18.1	19.5	16.6	12.3	7.0	1.0	0.1
	10%	WB	-3.1	-1.0	-1.3	1.5	7.8	11.9	14.0	12.9	9.9	5.0	-0.5	-1.7
		MCDB	-2.7	-0.7	-1.0	3.0	11.3	15.6	17.4	15.1	11.5	5.6	-0.2	-1.3
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.3	6.8	7.2	8.3	8.4	7.0	6.4	5.8	4.5	4.8	5.4
		MCDBR	5.7	4.8	5.4	9.0	12.5	12.5	12.2	9.6	7.4	5.0	4.1	5.3
		MCWBR	5.7	4.6	4.8	5.8	7.0	6.4	6.3	5.0	4.7	3.9	4.0	4.8
	5% WB	MCDBR	5.7	4.8	4.9	8.4	11.6	11.4	11.1	8.2	6.2	4.6	4.2	5.5
		MCWBR	5.7	4.8	4.4	5.6	6.6	5.9	6.1	4.7	4.3	3.9	4.2	5.2
Clear-Sky Solar Irradiance	taub	0.209	0.269	0.276	0.347	0.370	0.362	0.367	0.380	0.327	0.288	0.231	0.197	
	taud	2.035	2.168	2.235	2.089	2.269	2.391	2.415	2.354	2.490	2.391	2.013	1.882	
	Ebn at Noon	487	692	841	829	845	862	847	797	787	679	421	270	
	Edh at Noon	35	66	88	127	117	106	101	99	71	53	35	22	
All-Sky Solar Radiation	RadAvg	0.30	1.00	2.81	4.47	5.09	5.25	4.24	3.38	2.22	1.09	0.35	0.14	
	RadStd	0.04	0.12	0.26	0.33	0.48	0.54	0.33	0.43	0.20	0.09	0.06	0.01	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	+1.67	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page